

SHAORYANG, China

WMO: 577660

Lat: 27.1850N

Lon: 111.4535E

Elev: 310

StdP: 97.65

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.2	0.0	-8.6	1.9	5.4	-6.0	2.4	5.2	5.6	6.8	5.0	7.0	1.6	20	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.0	35.3	25.3	34.2	25.3	33.1	25.2	26.9	31.5	26.4	31.0	26.0	30.6	2.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.0	28.4	25.3	21.3	28.3	24.9	20.7	28.2	87.0	31.6	84.9	31.2	83.1	30.7	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	5.4	4.6	4.0	DB	-2.9	36.9	1.7	1.2	-4.2	37.7	-5.2	38.4	-6.1	39.0	-7.4	39.9
(5)				WB	-3.6	27.6	1.6	0.7	-4.8	28.1	-5.7	28.5	-6.6	28.8	-7.8	29.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.0	5.8	8.0	12.4	18.2	22.5	25.7	28.6	28.1	24.5	19.3	13.7	8.0
	DBStd	8.59	3.71	4.61	4.44	4.26	3.11	2.61	2.32	2.52	3.39	3.45	3.89	3.15
	HDD10.0	339	139	86	25	1	0	0	0	0	0	0	11	77
	HDD18.3	1423	388	289	191	54	3	0	0	0	2	29	146	321
	CDD10.0	3246	9	31	99	248	388	470	577	562	436	289	121	15
	CDD18.3	1288	0	1	6	51	133	220	319	303	188	60	7	0
	CDH23.3	12426	1	8	43	331	935	1942	3828	3387	1604	312	36	0
	CDH26.7	4984	0	1	8	86	286	671	1776	1486	601	66	5	0
Wind	WSAvg	1.6	1.5	1.6	1.6	1.7	1.6	1.6	1.8	1.8	1.8	1.6	1.5	1.5
Precipitation	PrecAvg	1303	55	81	107	185	202	199	104	107	63	89	74	39
	PrecMax	1804	129	155	197	376	470	387	283	241	233	208	187	83
	PrecMin	922	6	11	25	86	66	52	18	6	0	0	0	1
	PrecStd	219	30	39	39	63	93	87	63	65	53	63	51	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.4	24.4	27.5	31.5	33.0	34.1	36.5	36.9	35.1	31.0	27.0	19.6
		MCWB	14.2	16.8	20.2	21.8	23.4	25.7	25.6	25.2	24.4	21.0	19.2	12.3
	2%	DB	16.2	20.6	24.0	29.2	31.3	32.8	35.3	35.4	33.3	28.5	24.0	17.2
		MCWB	10.8	14.9	17.5	21.4	22.8	25.3	25.6	25.2	24.4	20.4	16.7	11.0
	5%	DB	13.5	17.4	21.5	27.1	29.8	31.6	34.3	34.1	31.7	26.6	21.9	15.2
		MCWB	9.2	12.8	16.2	20.5	22.6	25.1	25.4	25.3	23.9	19.1	15.6	9.8
	10%	DB	11.3	14.6	19.2	25.0	28.1	30.4	33.3	32.8	30.2	24.9	19.8	13.1
		MCWB	8.0	11.1	14.7	19.6	22.1	24.6	25.4	25.3	23.3	18.4	14.4	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	17.4	21.2	23.8	25.7	26.9	27.6	27.5	26.5	24.1	20.4	14.8
		MCDB	18.8	23.2	26.5	29.2	30.0	31.0	32.1	32.0	31.1	28.5	24.8	16.8
	2%	WB	11.7	15.5	18.9	22.6	24.8	26.2	26.9	26.8	25.6	22.2	18.4	12.4
		MCDB	14.5	20.0	22.6	27.4	28.6	30.5	31.8	31.5	30.4	25.3	22.0	15.4
	5%	WB	9.8	13.5	17.2	21.7	24.1	25.8	26.5	26.3	25.0	21.0	17.0	10.7
		MCDB	12.7	16.6	20.2	25.6	27.7	30.0	31.3	30.9	29.7	24.5	20.0	13.9
	10%	WB	8.2	11.6	15.5	20.5	23.2	25.4	26.1	25.9	24.4	20.0	15.6	9.5
		MCDB	10.9	14.1	18.2	23.8	26.4	29.2	30.8	30.4	28.9	23.4	18.5	12.4
Mean Daily Temperature Range	5% DB	MDBR	5.2	5.7	6.4	7.1	7.0	6.2	7.0	7.1	7.2	6.9	6.9	6.2
		MCDBR	10.6	11.6	11.3	11.0	10.1	8.2	9.0	9.2	9.6	10.1	10.9	11.4
		MCWBR	5.9	6.3	5.4	4.5	3.3	2.5	2.1	2.0	2.6	3.4	4.5	5.8
	5% WB	MCDBR	8.9	10.2	9.8	9.4	7.6	7.0	7.8	7.7	8.3	7.4	8.5	8.6
		MCWBR	5.4	6.2	5.7	4.9	3.6	2.7	2.4	2.3	2.6	3.4	4.3	4.8
Clear-Sky Solar Irradiance	taub	0.682	0.714	0.773	0.747	0.662	0.643	0.569	0.654	0.686	0.741	0.674	0.662	
	taud	1.497	1.476	1.420	1.509	1.703	1.759	1.957	1.724	1.630	1.478	1.564	1.549	
	Ebn at Noon	557	588	590	628	686	695	747	683	640	561	560	547	
	Edh at Noon	257	282	314	294	242	228	186	234	251	277	238	235	
All-Sky Solar Radiation	RadAvg	1.62	1.94	2.58	3.26	3.71	3.91	5.12	4.49	3.67	2.90	2.36	1.89	
	RadStd	0.35	0.53	0.33	0.45	0.59	0.56	0.57	0.35	0.39	0.42	0.39	0.45	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (6 neighbors)	+0.30	N/A	N/A	+0.59	N/A	N/A	N/A	N/A	+136	+74	

Nomenclature: See separate page